

A Look at **SQLly eXtensible PostgreSQL**

PostgreSQL, the world's most advanced open source database, has many cutting-edge features that empower its users. One of the important aspects of PostgreSQL functionality is its extensibility, which this article explores, with details on how the PostgreSQL functionality can be extended using SQL APIs. Readers with some background in databases will find the topic interesting.



In September 2012, the PostgreSQL Global Development Group released version 9.2 of PostgreSQL, which it claims to be "...the world's most advanced open source database." The release is packed with many features in the areas of performance improvements, SQL functionality, HA, replication, support for NoSQL data-types, etc.

PostgreSQL supports a lot of advanced features like Multi-Version Concurrency Control (MVCC), which allows non-blocking reads for higher transaction performance, SQL/MED for accessing external data—both

relational and flat files (CSV), object-oriented features like function overloading and table inheritance, table partitioning through table inheritance and constraint exclusion. It also supports streaming replication with hot standby, which provides high availability (HA) and increased read performance, XML and JSON data-types for NoSQL lovers, and the list continues.

Needless to say, PostgreSQL provides traditional features like ACID compliance, integrity constraints, backup and restore, triggers, stored procedures and the functions that are required for any relational database system.